1. **Revenue by product category.**

```
select product_category, round(SUM(o.quantity*unit_price*(1-discount/100.0)),2) as revenue_per_category
from        orders o
where       o.order_status = 'Completed'
group by    o.product_category
order by    revenue_per_category desc;
```

	product_category	revenue_per_category
1	Mobile Phones	6,942,060.65
2	Computing	6,596,221.7
3	Electronics	5,515,359.2
4	Home Appliances	4,024,234.05
5	Accessories	1,077,496.1

2. **Category revenue, cost, profit, and profit margin**

```
with category_aggregates  as (
        select o.product_category,
        SUM(quantity*unit_price * (1-discount/100.0))as category_revenue,
        sum(cost_price*quantity) as category_cost
        from orders o
        where o.order_status = 'Completed'
        group by o.product_category )
select  product_category,
        category_revenue,
        category_cost,
        category_revenue - category_cost as profit,
        round(100.0 * (category_revenue - category_cost)/ category_revenue ,2) as profit_margin
from category_aggregates
order by profit desc;
```

	product_category	category_revenue	category_cost	profit	profit_margin
1	Mobile Phones	6,942,060.65	6,213,300	728,760.65	10.5
2	Computing	6,596,221.7	5,896,200	700,021.7	10.61
3	Electronics	5,515,359.2	4,968,000	547,359.2	9.92
4	Home Appliances	4,024,234.05	3,566,200	458,034.05	11.38
5	Accessories	1,077,496.1	620,750	456,746.1	42.39

3. Trading Cities Vs Revenue Generated

```
select distinct c.city , round(SUM(o.quantity * unit_price *(1-discount/100.0)),2) as total_spent
from customers c
join orders o using (customer_id)
where o.order_status = 'Completed'
group by c.city
order by total_spent  desc
Limit 10;
```

	city	total_spent
1	Nairobi	9,581,980.35
2	Mombasa	3,161,889.6
3	Nakuru	2,391,328
4	Kiambu	1,653,531
5	Meru	1,640,791
6	Eldoret	1,448,789.3
7	Kisumu	1,374,276.65
8	Thika	1,163,583.7
9	Nyeri	994,102.2
10	Machakos	745,099.9

4. Average Customer Spending By Segment

```
select c.customer_segment ,
       count(distinct c.customer_id) as customers,
       count (distinct o.customer_id) as buyers,
       sum(o.quantity * unit_price *(1-discount/100.0)) as revenue,
       round(coalesce (sum(o.quantity * unit_price *(1-discount/100.0)),0)/count(distinct c.customer_id ),2) as avg_total_spent
from   customers c
left join  orders o  on c.customer_id = o.customer_id
           and o.order_status = 'Completed'
group by   c.customer_segment
order by   avg_total_spent  desc;
```

	customer_segment	customers	buyers	revenue	avg_total_spent
1	VIP	19	19	4,811,785.75	253,251.88
2	Corporate	29	25	4,948,967.75	170,654.06
3	Retail	102	88	14,394,618.2	141,123.71

5. Monthly Sales Summary

```
create or replace function get_customer_revenue(p_customer_id int)
        returns numeric
        language plpgsql
        as $$
        declare
            v_result numeric;
        begin
            select sum(quantity*unit_price* (1-discount/100.0))
            into v_result
            from orders o
            where customer_id = p_customer_id
                and order_status = 'Completed';
            return coalesce(round(v_result,2),0);
        end;
$$;
```

	month	123 orders	123 revenue	123 cost	123 profit
1	2025-01-01	10	434,902.7	414,700	20,202.7
2	2025-02-01	13	369,565.6	316,600	52,965.6
3	2025-03-01	20	735,247.55	649,750	85,497.55
4	2025-04-01	16	632,642.9	565,900	66,742.9
5	2025-05-01	18	682,628.6	636,150	46,478.6
6	2025-06-01	18	772,390.35	662,800	109,590.35
7	2025-07-01	24	1,235,096.25	1,081,450	153,646.25
8	2025-08-01	26	1,056,881.4	930,300	126,581.4
9	2025-09-01	26	968,117.95	850,300	117,817.95
10	2025-10-01	31	1,139,331.9	990,350	148,981.9
11	2025-11-01	42	1,269,360.9	1,139,450	129,910.9
12	2025-12-01	59	2,027,120.6	1,792,900	234,220.6
13	2026-01-01	14	322,432.95	265,700	56,732.95
14	2026-02-01	28	762,338.85	663,800	98,538.85
15	2026-03-01	25	1,013,348.2	958,750	54,598.2
16	2026-04-01	40	1,654,863.3	1,436,500	218,363.3
17	2026-05-01	31	1,178,459.1	1,017,500	160,959.1
18	2026-06-01	43	2,038,372.05	1,794,650	243,722.05
19	2026-07-01	44	1,527,791.5	1,335,000	192,791.5
20	2026-08-01	54	2,242,033.9	1,968,200	273,833.9
21	2026-09-01	52	2,092,445.15	1,793,700	298,745.15